

Taledge Enterprise AI Interview Pipeline

Architectural Overview

This document details the exact sequence diagrams, payload schemas, and sub-system architectures that power the Taledge Dual-Track AI Interview Engine. The system leverages an asynchronous, bi-directional streaming architecture with real-time semantic processing, advanced adversarial LLM routing, and heuristic proctoring matrices.

1. System Initialization & State Hydration

Before the WebRTC or WebSocket layers establish a connection, the `EngineController` hydrates a strict execution context. This prevents hallucination and binds the LLM to a deterministic evaluation path.

1.1 The Multi-Dimensional Context Payload

The frontend compiles an aggregated, token-optimized JSON structure containing parsed ATS data, Job Description (JD) Vector Embeddings, and Academic historical markers.

```
{
  "session_id": "vtx_99812a_b7",
  "execution_mode": "adversarial_tech",
  "candidate_matrix": {
    "identity_hash": "2f4b9d0a",
    "target_role": "Senior Cloud Solutions Architect",
    "skill_graph_embeddings": [0.012, -0.045, 0.992, ...],
```

```
"verified_tech_stack": {
  "primary": ["Kubernetes", "AWS EKS", "Go"],
  "secondary": ["PostgreSQL", "Terraform"]
},
"impact_claims": [
  {
    "domain": "Distributed Systems",
    "claim": "Engineered a zero-downtime
deployment pipeline scaling to 50k RPS.",
    "validation_flag": "REQUIRES_DEEP_PROBING"
  }
]
},
"constraints": {
  "max_turn_latency_ms": 1200,
  "cognitive_load_multiplier": 1.5
}
}
```

2. The Deterministic LLM Persona Routing

The "Brain" of the system relies on dynamic system-prompt injection. Depending on the `execution_mode`, the context is routed to specialized sub-models.

2.1 The Elite Technical Adversary (Stage 1)

Instead of a generic system prompt, the LLM is bound by a strict adversarial framework designed to induce and measure cognitive load.

SYSTEM PROMPT BINDING:

[Role]: Elite Principal Engineer & Hostile System

Architect.

[Language Parameters]: Multilingual continuous (English/Hindi/Hinglish) with auto-detection.

[Execution Directives]:

1. TARGET CLAIMS: Isolate the candidate's claim: "zero-downtime deployment pipeline scaling to 50k RPS."
2. STRESS VECTOR: Ask them how their architecture handles a split-brain network partition in the Redis caching layer during a deploy.
3. COGNITIVE LOAD: If they answer correctly, immediately compound the load by introducing a sudden IOPS bottleneck on the primary PostgreSQL shard.
4. OUTPUT FORMAT: Maximum 45 words. High density. No pleasantries. Probe edge cases strictly. Do not validate their answers.

2.2 The Psychometric DNLA Analyst (Stage 2)

SYSTEM PROMPT BINDING:

[Role]: Clinical Behavioural Psychologist & HR Director.

[Execution Directives]:

1. BEHAVIOURAL TRAPPING: Bypass standard STAR methodology. Induce cognitive dissonance by asking them to defend a time they knowingly bypassed protocol to meet a deadline.
2. LINGUISTIC MAPPING: Track Pronoun Indexing (I vs We). If 'We' is used for success and 'They' for failure, trigger the [Accountability Deficit] flag.
3. EMOTIONAL REGULATION: Apply mild conversational

pressure to measure baseline defensive mechanism triggers.

3. The Bi-Directional Streaming Pipeline

The conversational loop is not a standard HTTP Request/Response model. It uses an ultra-low latency continuous stream.

3.1 Step-by-Step Subsystem Execution

1. **Auditory Ingestion (STT Array):** The candidate speaks. The browser captures audio chunks via the MediaRecorder API, feeding it into a VAD (Voice Activity Detection) buffer. Once a pause of `700ms` is detected, the transcribing Webkit API fires the finalized payload.
2. **Pre-Processing & Sanitization:** The raw text is passed through a semantic normalizer to strip filler words ("um", "uh") and extract core intent, reducing token bloat before LLM transmission.
3. **LLM Inference Generation:** The backend streams the normalized history to the LLM router (OpenRouter fallback cluster). The generation is parameterized with `temperature: 0.6` and `top_p: 0.8` to ensure high determinism and low hallucination.
4. **Chunked TTS Streaming:** To eliminate waiting latency, the LLM's response is chunked by sentence delimiters (`.` , `?`). As soon as the first sentence is generated, it is pushed to the client via Server-Sent Events (SSE) and immediately vocalized by the `SpeechSynthesis` Web API while the LLM continues generating the rest in the background.

4. Multi-Vector Security & Proctoring Environment

To guarantee assessment integrity, the interview environment runs inside a heavily sandboxed, heuristic monitoring wrapper.

4.1 Environmental Heuristics

- **Context-Switch Deterrence:** A strict `visibilitychange` listener is bound to the document root. If `document.hidden` returns `true` , a Level 1 Violation is recorded. At 3 violations, the WebSocket is forcefully severed, and the session is burned.
- **Micro-Expression Latency Tracking:** The system calculates the exact millisecond delta between the AI finishing its speech and the candidate beginning theirs. A latency > 4000ms triggers a `HIGH_HESITATION` flag, heavily weighting the confidence score downwards.
- **Copy-Paste Nullification:** All standard clipboard events (`keydown` for `Meta+C/V` , `contextmenu`) are intercepted and `preventDefault()` is called.

5. Session Termination & State Output

Upon termination (either via natural token limit exhaustion or a security breach), the `EngineController` compiles the entire transcript, the

latency metadata, and the proctoring logs into a finalized

`AssessmentLedger` . This ledger is then placed into a high-throughput queue (e.g., Redis/Kafka) where the asynchronous Scoring Engine (`/api/generate-fit-score`) takes over to calculate the final Fit Score.